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First-Order Gradient Descent Methods

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First-Order Descent

- First-order methods work with steepest-descent directions.

- Modifications to basic form of steepest-descent:
  - Need to reduce step sizes with algorithm progression.
  - Need a way of avoiding local optima.
  - Need to address widely varying slopes with respect to different weight parameters.

Learning Rate Decay

- Initial learning rates should be high but reduce over time.

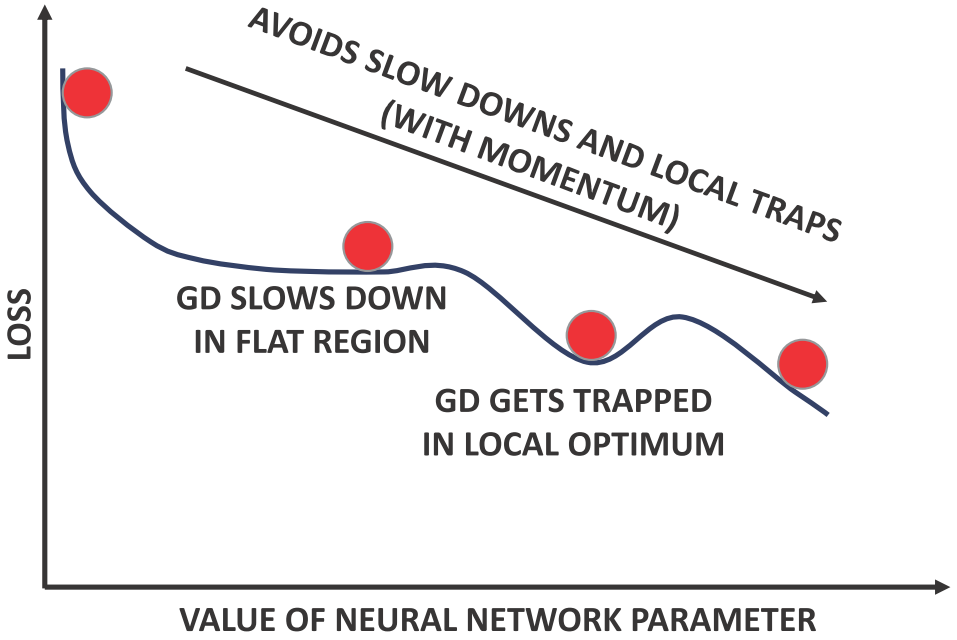
- The two most common decay functions are *exponential decay* and *inverse decay*.

- The learning rate α_t can be expressed in terms of the initial decay rate α_0 and epoch t as follows:

$$\alpha_t = \alpha_0 \exp(-k \cdot t) \quad \text{[Exponential Decay]}$$
$$\alpha_t = \frac{\alpha_0}{1 + k \cdot t} \quad \text{[Inverse Decay]}$$

  The parameter k controls the rate of the decay.

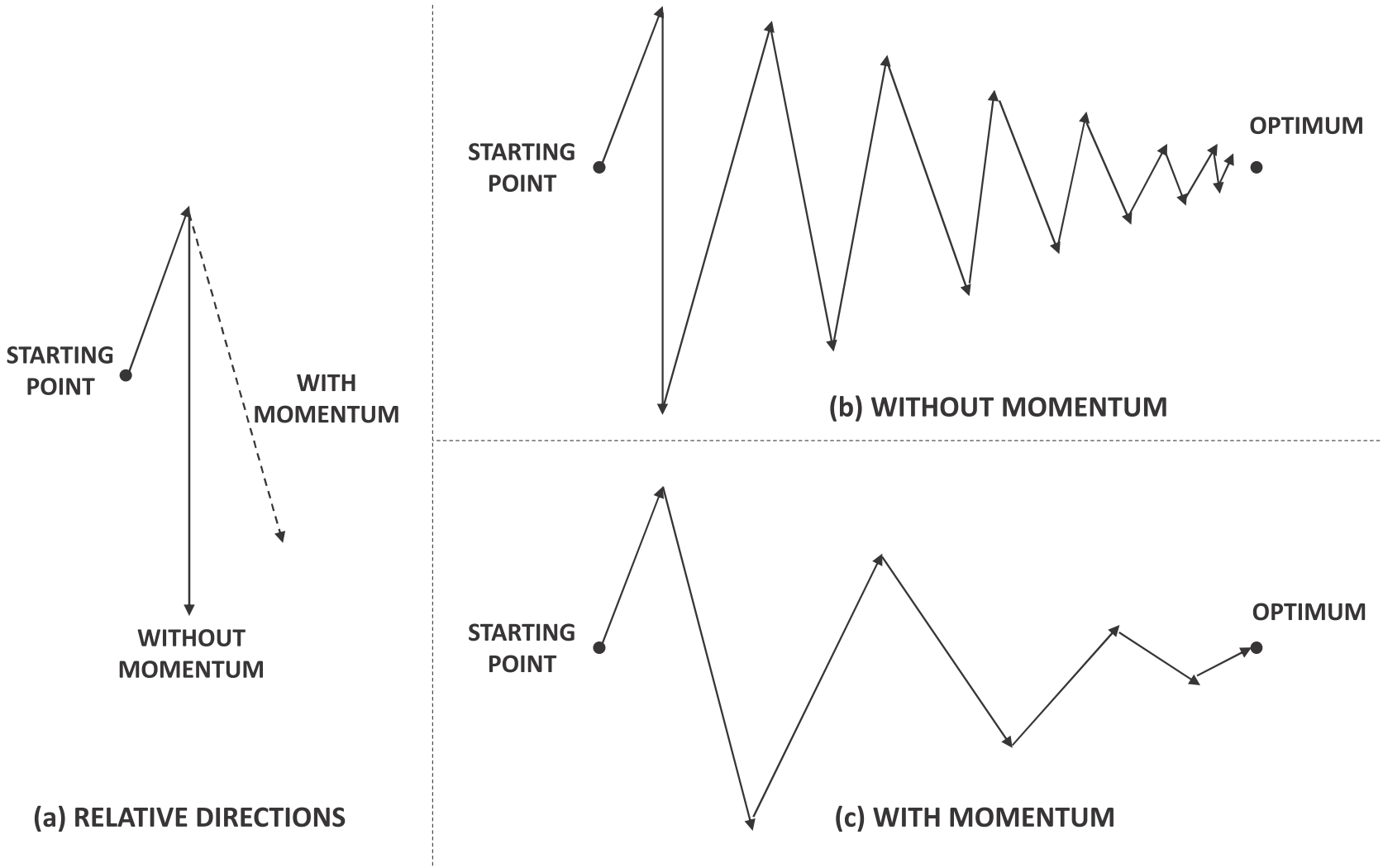
Momentum Methods: Marble Rolling Down Hill



- Use a *friction parameter* $\beta \in (0, 1)$ to gain speed in direction of movement.

$$\overline{V} \Leftarrow \beta \overline{V} - \alpha \frac{\partial L}{\partial \overline{W}}; \quad \overline{W} \Leftarrow \overline{W} + \overline{V}$$

Avoiding Zig-Zagging with Momentum



Nesterov Momentum

- Modification of the traditional momentum method in which *the gradients are computed at a point that would be reached after executing a β-discounted version of the previous step again*.

- Compute at a point reached using only the momentum portion of the current update:

$$\overline{V} \Leftarrow \underbrace{\beta\overline{V}}_{\text{Momentum}} - \alpha\frac{\partial L(\overline{W}+\beta\overline{V})}{\partial\overline{W}}; \quad \overline{W} \Leftarrow \overline{W}+\overline{V}$$

- Put on the brakes as the marble reaches near bottom of hill.

- Nesterov momentum should always be used with mini-batch SGD (rather than SGD).

AdaGrad

- *Aggregate* squared magnitude of ith partial derivative in A_i.
- The square-root of A_i is proportional to the root-mean-square slope.
  - The absolute value will increase over time.

$$A_i \Leftarrow A_i + \left(\frac{\partial L}{\partial w_i}\right)^2 \quad \forall i \tag{15}$$

- The update for the ith parameter w_i is as follows:

$$w_i \Leftarrow w_i - \frac{\alpha}{\sqrt{A_i}} \left(\frac{\partial L}{\partial w_i}\right); \quad \forall i \tag{16}$$

- Use $\sqrt{A_i + \epsilon}$ in the denominator to avoid ill-conditioning.

AdaGrad Intuition

- Scaling the derivative inversely with $\sqrt{A_i}$ encourages faster *relative* movements along gently sloping directions.
  - Absolute movements tend to slow down prematurely.
  - Scaling parameters use stale values.

RMSProp

- The RMSProp algorithm uses *exponential smoothing* with parameter $\rho \in (0, 1)$ in the relative estimations of the gradients.
  - Absolute magnitudes of scaling factors do not grow with time.
  - Problem of staleness is ameliorated.

$$A_i \Leftarrow \rho A_i + (1 - \rho) \left(\frac{\partial L}{\partial w_i} \right)^2 \quad \forall i \tag{17}$$

$$w_i \Leftarrow w_i - \frac{\alpha}{\sqrt{A_i}} \left(\frac{\partial L}{\partial w_i} \right); \quad \forall i$$

- Use $\sqrt{A_i + \epsilon}$ to avoid ill-conditioning.

RMSProp with Nesterov Momentum

- Possible to combine RMSProp with Nesterov Momentum

$$v_i \Leftarrow \beta v_i - \frac{\alpha}{\sqrt{A_i}} \left(\frac{\partial L(\overline{W} + \beta \overline{V})}{\partial w_i} \right); \quad w_i \Leftarrow w_i + v_i \quad \forall i$$

- Maintenance of A_i is done with shifted gradients as well.

$$A_i \Leftarrow \rho A_i + (1 - \rho) \left(\frac{\partial L(\overline{W} + \beta \overline{V})}{\partial w_i} \right)^2 \quad \forall i \qquad (18)$$

AdaDelta and Adam

- Both methods derive intuition from RMSProp
  - AdaDelta track of an exponentially smoothed value of the *incremental changes* of weights Δw_i in previous iterations to decide parameter-specific learning rate.
  - Adam keeps track of exponentially smoothed gradients from previous iterations (in addition to normalizing like RMSProp).
- Adam is extremely popular method.